

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

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STEP0: CREATE DATABASE AND USE

```
CREATE DATABASE IF NOT EXISTS DDL_DB;
```

```
USE DDL_DB;
```

OUTPUT:

✓	2	19:41:29	USE DDL_DB
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Step1: Create Table

```
1 CREATE TABLE bank_transactions (  
2     txn_id INT PRIMARY KEY,  
3     customer_name VARCHAR(50),  
4     branch_name VARCHAR(50),  
5     transaction_type VARCHAR(20),  
6     amount DECIMAL(10,2),  
7     transaction_date DATE  
8 );
```

Output

Action Output

#	Time	Action	Message
✓ 1	16:27:06	CREATE TABLE bank_transactions (txn_id INT PRIMARY KE...	0 row(s) affected

Step2: ALTER TABLE ADD COLUMN

```

9 • ALTER TABLE bank_transactions
10 ADD account_no VARCHAR(20);
11

```

Output

Action Output

#	Time	Action
1	16:27:06	CREATE TABLE bank_transactions (txn_id INT PRIMARY KE...
2	16:36:34	ALTER TABLE bank_transactions ADD account_no VARCHAR(.

STEP3: ALTER TABLE MODIFY COLUMN

```

11 • ALTER TABLE bank_transactions
12 MODIFY customer_name VARCHAR(100);

```

Output

Action Output

#	Time	Action
1	16:27:06	CREATE TABLE bank_transactions (txn_id INT PRIMARY KE...
2	16:36:34	ALTER TABLE bank_transactions ADD account_no VARCHAR(.
3	16:38:36	ALTER TABLE bank_transactions MODIFY customer_name VA..

STEP4:RENAME TABLE

```

14 • RENAME TABLE bank_transactions TO customer_transactions;
15

```

Output

Action Output

#	Time	Action	Message
1	16:46:17	RENAME bank_transactions TO customer_transactions	Error C
2	16:46:51	RENAME bank_transactions TO customer_transactions	Error C
3	16:47:51	ALTER TABLE bank_transactions MODIFY customer_name VA...	0 row(s)
4	16:47:54	RENAME bank_transactions TO customer_transactions	Error C
5	16:48:43	RENAME TABLE bank_transactions TO customer_transactions	0 row(s)

STEP5:TRUNCATE TABLE

```
TRUNCATE TABLE customer_transactions;
```

OUTPUT:

```
8 19:42:54 TRUNCATE TABLE customer_transactions
```

STEP6 - DROP TABLE

```
DROP TABLE IF EXISTS bank_backup;
```

OUTPUT:

⚠	12	19:50:11	DROP TABLE IF EXISTS bank_backup	0 row(s) affected, 1 warning
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STEP7: DML (DATA MANIPULATION LANGUAGE)

INSERT SAMPLE DATA

```
2  CREATE TABLE customers_transactions (  
3      txn_id INT PRIMARY KEY,  
4      customer_name VARCHAR(50),  
5      branch_name VARCHAR(50),  
6      transaction_type VARCHAR(20),  
7      amount DECIMAL(10,2),  
8      transaction_date DATE  
9  );  
10  
11  INSERT INTO customers_transactions VALUES  
12      (101, 'Vaishnavi', 'Hyderabad', 'Deposit', 2000, '2024-01-05'),  
13      (102, 'Likhitha', 'Hyderabad', 'Withdrawal', 2000, '2024-01-07'),  
14      (103, 'Kushal', 'Hyderabad', 'Deposit', 12000, '2024-01-08'),  
15      (104, 'Sai', 'Guntur', 'Deposit', 10000, '2024-01-10'),  
16      (105, 'Sushma', 'Hyderabad', 'Withdrawal', 8000, '2024-01-11'),  
17      (106, 'Archana', 'Vijayawada', 'Deposit', 1000, '2024-01-12'),  
18      (107, 'Dikshitha', 'Vijayawada', 'Withdrawal', 2000, '2024-01-13'),  
19      (108, 'Divya', 'Guntur', 'Deposit', 5000, '2024-01-13'),  
20      (109, 'Sri', 'Guntur', 'Withdrawal', 15000, '2024-01-14'),  
21      (110, 'Akhila', 'Hyderabad', 'Deposit', 1000, '2024-01-15');  
22
```

Output

Action Output

#	Time	Action	Message
✓ 2	09:15:53	CREATE TABLE customers_transactions (txn_id INT PRIMARY KEY, customer_name...	0 row(s) affected

STEP8:INSERT

```
23  INSERT INTO customers_transactions  
24  VALUES  
25      (111, 'venu', 'hyderabad', 'Deposit', 10000, '2024-01-17');  
26  
27
```

Output

Action Output

#	Time	Action	Message
✓ 3	09:16:19	INSERT INTO customers_transactions VALUES (101, 'Vai...	10 row(s)

Step9: update

```
28 • UPDATE customers_transactions
29   SET amount = 1000
30   WHERE txn_id = 105;
31
```

Context Hel

Output

Action Output

#	Time	Action	Message
✓ 4	09:20:12	INSERT INTO customers_transactions VALUES (111, v...	1 row(s) affected
✓ 5	09:22:01	UPDATE customers_transactions SET amount = 1000 W...	1 row(s) affected Rows matched: 1 Changed:

Step10:DELETE

```
31 • DELETE FROM customers_transactions
32   WHERE txn_id = 111;
33
```

Output

Action Output

#	Time	Action	Mess
✓ 5	09:22:01	UPDATE customers_transactions SET amount = 1000 W...	1 row
✓ 6	09:25:34	DELETE FROM customers_transactions WHERE txn_id = ...	1 row

STEP11: SELECT

```
34 • SELECT *
35   FROM customers_transactions;
36
37
```

Limit to 1000 rows

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Vaishnavi	Hyderabad	Deposit	2000.00	2024-01-05
102	Likhitha	Hyderabad	Withdrawal	2000.00	2024-01-07
103	Kushal	Hyderabad	Deposit	12000.00	2024-01-08
104	Sai	Guntur	Deposit	10000.00	2024-01-10
105	Sushma	Hyderabad	Withdrawal	1000.00	2024-01-11
106	Archana	Vijayawada	Deposit	1000.00	2024-01-12
107	Dikshitha	Vijayawada	Withdrawal	2000.00	2024-01-13
108	Divya	Guntur	Deposit	5000.00	2024-01-13
109	Sri	Guntur	Withdrawal	15000.00	2024-01-14
110	Akhila	Hyderabad	Deposit	1000.00	2024-01-15

customers_transactions 19 x Apply

Output

Action Output

#	Time	Action	Message
✗ 8	09:27:06	SELECT * FROM customers_transactions LIMIT 0, 1000	Error Code: 1146. Table 'ddl_db.customers_' doesn't e
✓ 9	09:27:22	SELECT * FROM customers_transactions LIMIT 0, 1000	10 row(s) returned

STEP12: SELECT WITH WHERE

35 FROM customers_transactions;
36 • SELECT *
37 FROM customers_transactions
38 WHERE transaction_type = 'Deposit';
39
40

Result Grid

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	101	Vaishnavi	Hyderabad	Deposit	2000.00	2024-01-05
	103	Kushal	Hyderabad	Deposit	12000.00	2024-01-08
	104	Sai	Guntur	Deposit	10000.00	2024-01-10
	106	Archana	Vijayawada	Deposit	1000.00	2024-01-12
	108	Divya	Guntur	Deposit	5000.00	2024-01-13
	110	Akhila	Hyderabad	Deposit	1000.00	2024-01-15
*	NULL	NULL	NULL	NULL	NULL	NULL

customers_transactions20

Output

Action Output

#	Time	Action	Message
9	09:27:22	SELECT * FROM customers_transactions LIMIT 0, 1000	10 row(s) returned
10	09:30:31	SELECT * FROM customers_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned

STEP13: SELECT WITH ORDER BY

40 • SELECT *
41 FROM customers_transactions
42 ORDER BY amount DESC;
43

Result Grid

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	109	Sri	Guntur	Withdrawal	15000.00	2024-01-14
	103	Kushal	Hyderabad	Deposit	12000.00	2024-01-08
	104	Sai	Guntur	Deposit	10000.00	2024-01-10
	108	Divya	Guntur	Deposit	5000.00	2024-01-13
	101	Vaishnavi	Hyderabad	Deposit	2000.00	2024-01-05
	102	Likitha	Hyderabad	Withdrawal	2000.00	2024-01-07
	107	Dikshitha	Vijayawada	Withdrawal	2000.00	2024-01-13
	105	Sushma	Hyderabad	Withdrawal	1000.00	2024-01-11
	106	Archana	Vijayawada	Deposit	1000.00	2024-01-12
	110	Akhila	Hyderabad	Deposit	1000.00	2024-01-15

customers_transactions21

Output

Action Output

#	Time	Action	Message
10	09:30:31	SELECT * FROM customers_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned
11	09:32:04	SELECT * FROM customers_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned

STEP14: DCL (DATA CONTROL LANGUAGE)

GRANT ALL

```
CREATE USER 'BranchManager'@'localhost'
IDENTIFIED BY 'Branch@123';
GRANT ALL PRIVILEGES
ON ddl_db.customers_transactions
To 'BranchManager'@'localhost';
SHOW GRANTS FOR 'BranchManager'@'localhost';
```

OUTPUT:

```
72 15:48:57 GRANT ALL PRIVILEGES ON ddl_db.customers... 0 row(s) affected
```

STEP15: REVOKE ALL

```
REVOKE ALL PRIVILEGES
ON ddl_db.customers_transactions
FROM 'BranchManager'@'localhost';
```

OUTPUT:

```
74 15:49:33 REVOKE ALL PRIVILEGES ON ddl_db.customer... 0 row(s) affected
```

Step16:TCL Commit

```
UPDATE customer_transactions
SET amount=6000
WHERE txn_id=101;
```

```
COMMIT;
```

OUTPUT:

```
16 09:47:59 COMMIT
```

Step17: savepoint

```
54 COMMIT;
55 SAVEPOINT Before_Update;
```

```
56
```

```
--
```

Output

Action Output

	#	Time	Action	
✓	16	09:47:59	COMMIT	0
✓	17	09:48:38	SAVEPOINT Before_Update	0

Step18:rollback to savepoint

```
UPDATE customer_transactions  
SET amount=99999  
WHERE txn_id=102;  
ROLLBACK TO Before_Update;
```

OUTPUT:

```
✓ 84 15:55:57 UPDATE customer_transactions SET amount=99...
```